ALICE L WALTON SCHOOL OF MEDICINE

DNA gyrase and Folic acid inhibitors– 2 of 2: Fluoroquinolones

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Adapted from: Fluoroquinolones

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of DNA Gyrase Inhibitors and Folic acid synthesis inhibitors.
2. Compare and contrast the pharmacokinetics of DNA Gyrase Inhibitors an
3. Recognize the common adverse reactions associated with DNA Gyrase and Folic acid synthesis inhibitors.
4. Identify the common drug interactions associated with DNA Gyrase and Folic acid synthesis inhibitors.

CASE CONNECTION

"I can't believe I've got an infection. I've got too much to do! And it's my 40th birthday!" exclaims VF while holding her side with one hand. You've known VF for many years; she must feel bad to have taken time to get checked out, because she is usually a go-go-go kind of person. She has had flank pain, pain with urination, frequent urination, and fever with occasional nausea. Her temperature is 100°F. She has left flank percussion tenderness and an otherwise normal exam.

How will you explain this illness to VF? Consider your answer as you read, and we'll revisit VF at the end of the brick.

GO TO CONCLUSION ↓

What Are Fluoroquinolones?

Fluoroquinolones (FQs) are a group of antibiotics that target the enzyme DNA gyrase and its sister enzyme, topoisomerase IV, to varying degrees. These antibiotics are a crucial part of antibiotic therapy worldwide and have held this role for decades. From relatively simple infections to the treatment of highly complex, life-threate of modern health care.

Fluoroquinolones get their name for a pretty obvious reason—they are a fluorinated form of quinolone, their core component (Figure 1). Show below are the drugs ciprofloxacin, which is commonly prescribed, and nalidixic acid, an original quinolone that is no longer on the market.

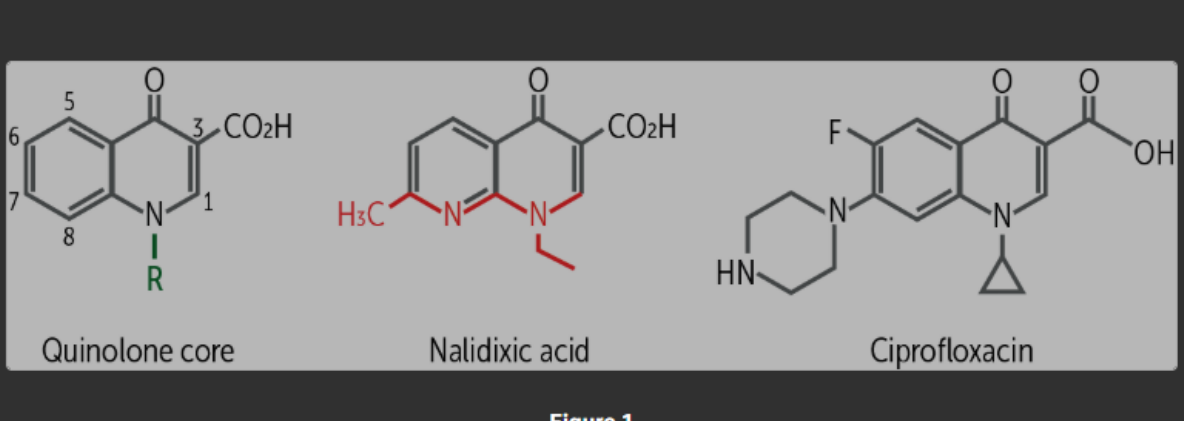
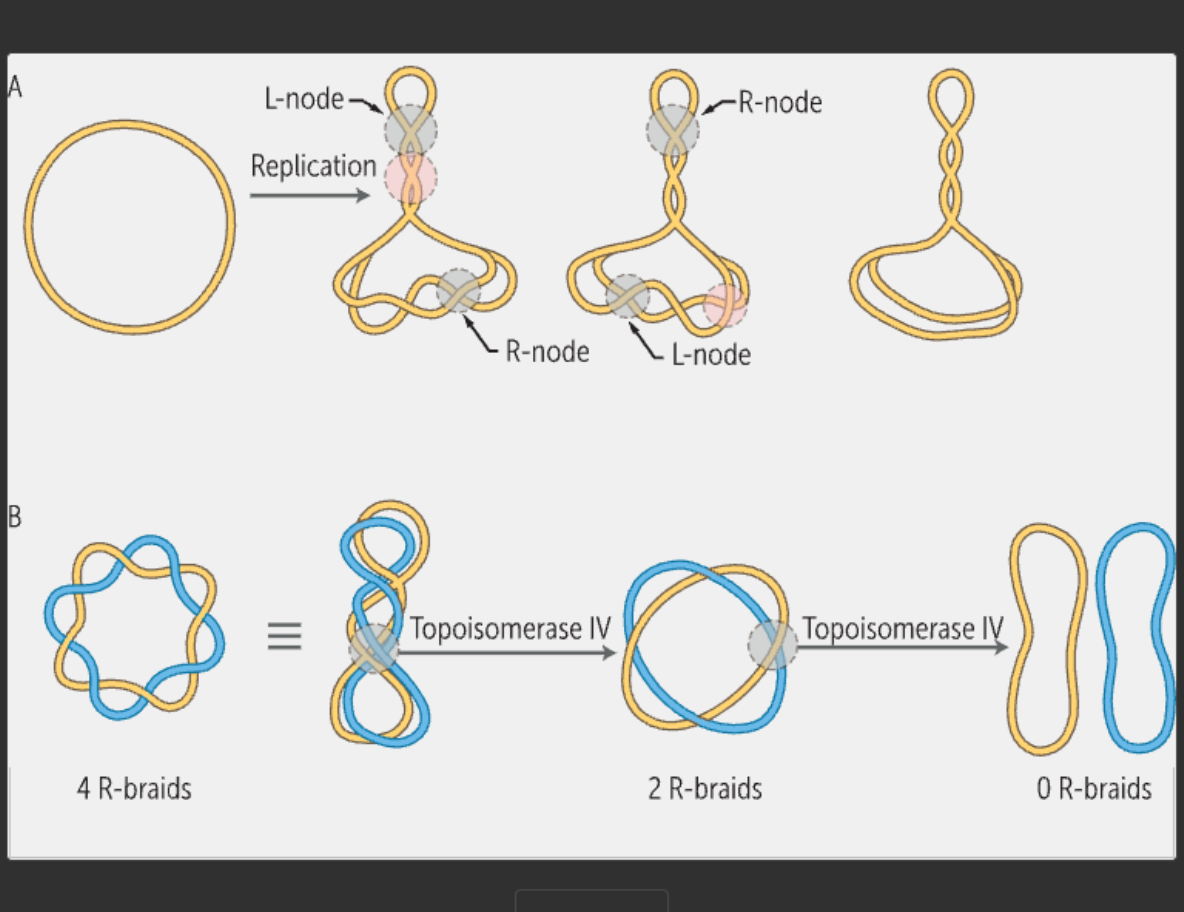


Figure 1

Credit: ©ScholarRx

What Is the Mechanism of Action of Fluoroquinolones? (LO#1)

All FQs do effectively the same thing: inhibit the type II topoisomerase DNA gyrase and topoisomerase IV with variable specificity within the class. Visualize the DNA double helix. Normally, DNA twists clockwise (Like you turn your steering wheel when making a right turn). If the two strands were pulled apart, the helix would wind tighter and tighter, creating what is called a positive supercoil. Now imagine taking a piece of yarn and twisting it; it gets tighter at first but eventually knots around itself. This is exactly what DNA would do during tran isomerases: The two enzymes clip the DNA downstream of DNA-reading machinery, allowing it to untwist and relax (Figure 2, A).



QUIZ

Figure 2

Credit: ©ScholarRx

Both DNA gyrase and topoisomerase IV have two types of domains: a ligase domain and a nuclease domain. The nuclease domain nicks a DNA strand, allowing the DNA to igase domain ligates (or repairs) this nick. DNA gyrase can uniquely create negative supercoils, which help prepare for further tension formed during DNA replication. Topoisomerase IV, on the other hand, can relax negative supercoils, a role shared by other topoisomerases, and plays a role in decatenation, that is, separating two completed bacterial chromosomes by passing one strand through the other (Figure 2, B).

What unique roles do DNA gyrase and topoisomerase IV have?

FQs target and inhibit the ligase domain of type II topoisomerases, allowing the nuclease to continue nicking the DNA while removing the repair mechanism—effectively making these crucial enzymes into DNA shredders. As a result, DNA replication is halted, causing bacterial cell death, which explains the bactericidal nature of FQs (Figure 3).

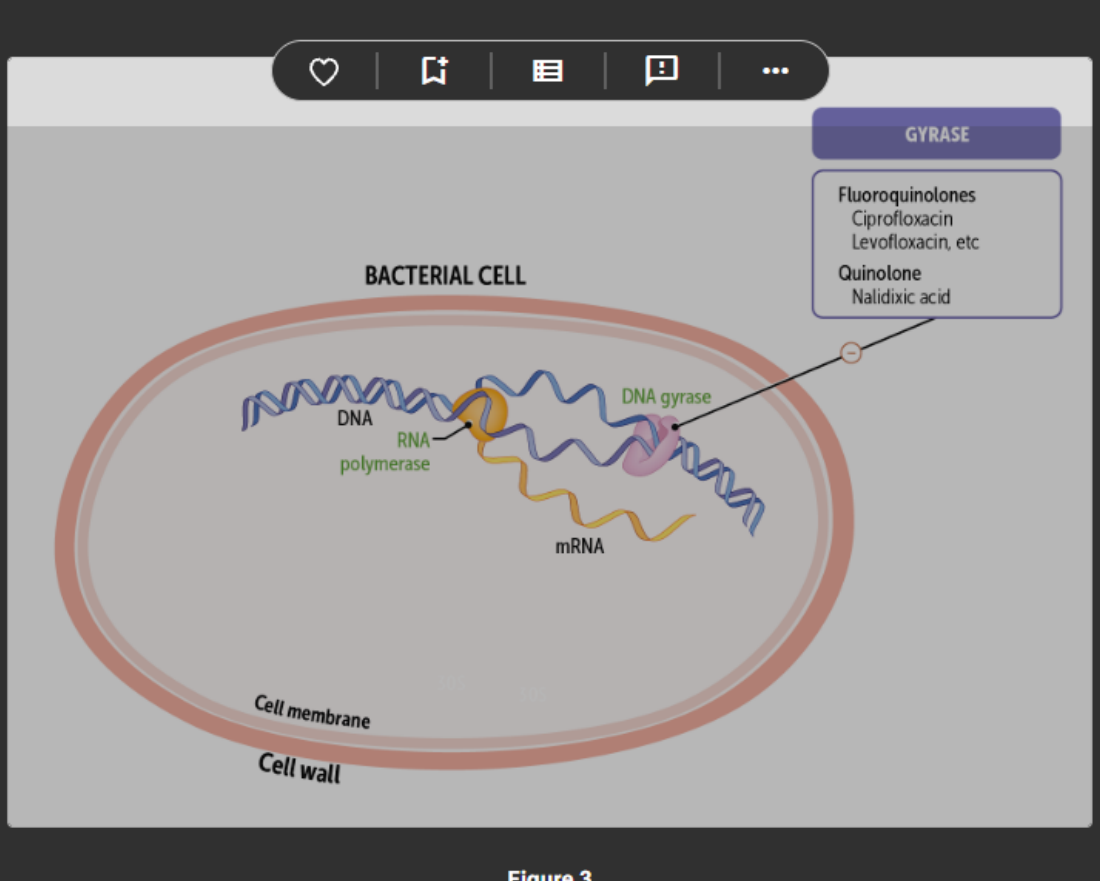


Figure 3

Credit: ©ScholarRx

The efficiency of these medications can be further increased by using synergistic medications together. For example, β-lactam antibiotics inhibit bacterial cell wall synthesis, allowing FQs to have an easier time entering bacterial cells to reach their target enzymes.

How Does Resistance to Fluoroquinolones Develop? (LO#1)

What are the two main mechanisms of bacterial resistance against FQs?

Like with every antibiotic, bacterial resistance is a constant concern for FQs, especially given their overuse. In general, there are two main ways that resistance may develop to an antibiotic:

- Resistance via active transport out of bacterial cells through efflux pumps
- Alteration of the target binding site (topoisomerase IV or DNA gyrase subunits) by chromosomal mutation, which may be chromosomal or integrated via plasmids (little loops of genetic code acquired from nearby bacterial cells)

Resistance to FQs is unfortunately widespread between both gram-positive and gram-negative bacteria. Among gram-positive organisms, methicillin-resistant *Staphylococcus aureus* (MRSA) is the most frequent culprit, while *Pseudomonas aeruginosa* is the most commonly resistant gram-negative bacterium. More recently, outbreaks of more resistant gram-negative organisms, such as *Neisseria gonorrhoeae* and *Campylobacter jejuni* have developed. Once one bacterium becomes re chanisms via plasmids may quickly protect a large population.

How Do We Choose the Right Fluoroquinolone? (LO#1)

No formal grouping of the fluoroquinolones exists, however, some are termed *respiratory fluoroquinolones.* This group of fluoroquinolones includes the medications: levofloxacin, moxifloxacin, and gatifloxacin (only available in opthalmic form in US). Levofloxacin is the most common of the respiratory fluoroquinolones because it has better *pseudomonas* coverage and lesser side-effect profile than moxifloxacin and gatifloxacin. They are termed *respiratory fluoroquinolones* because of their coverage against *S. pneumoniae*, but we will discuss their coverage in a later brick.

The other important fluoroquinolones include: ciprofloxacin, norfloxacin, and ofloxacin. Ciprofloxacin is the most effective when given systemically, and therefore, the most common of this group. Notably, norfloxacin has been withdrawn from the US market but is used elsewhere.

In general, FQs can also be split by where they're used. While ciprofloxacin is rarely used for respiratory infections, levofloxacin and moxifloxacin are key antibiotics for respiratory infections. Their use in respiratory infections (above the diaphragm) separates them from other systemic FQs, such as ciprofloxacin, used for other systemic infections (below the diaphragm).

Which fluoroquinolones are key antibiotics for respiratory infections?

How Are Fluoroquinolones Metabolized? (LO#2)

So, how do FQs get to where they need to go? First, know that the process differs based on how the FQ was dosed, either orally, parenterally, or topically (via cream or ophthalmic). While many FQs can act locally, some are able to have a systemic effect.

Distribution and Bioavailability (LO#4)

When taken orally, FQs are quickly absorbed through the gut wall, pass the hepatic circulation, and enter the blood to reach the systemic circulation. FQs as a class have a very high volume of distribution, and concentrations in some tissues surpass the peak serum concentration.

While food intake overall has little effect on FQ absorption, certain drug/drug classes can decrease the absorption of FQs. Divalent or trivalent cations such as aluminum, magnesium, calcium found in antacids can chelate with FQs and decrease absorption. Other agents such as iron and mineral supplements, and multivitamins also d eractions, FQs are given 4-6 hours after these medications.

What common items interfere with FQ metabolism?

Lastly, FQs, particularly ciprofloxacin, inhibit cytochrome P450 isozyme 1A2, which is also used in the elimination of theophylline and caffeine.

CLINICAL CORRELATION

As FQs inhibit cytochrome P450 enzymes, dosing of medications that are metabolized by this enzyme should be adjusted. Theophylline, for example, should have its dose significantly decreased if a patient starts ciprofloxacin therapy.

What Adverse E s Cause? (LO#3)

The apparently dangerous side effects of FQs have been a major topic of discussion in recent years, with a number of updated alerts by the US Food and Drug Administration. The most common side effects involve divergence from bacterial homeostasis. Lastly, disturbance of normal vaginal flora may lead to increased risk of candida vaginitis. If a patient experiences an adverse reaction, discontinue the FQ.

Gastrointestinal

The most common adverse effect with FQ use is transient or mild **GI upset**. Disturbances in the gut microbiome, or gut dysbiosis, lead to the classic GI side effects of nausea, vomiting, and diarrhea. A lack of competition by normal gut flora may allow *Clostridioides difficile* to proliferate, leading to overgrowth and *C difficile* colitis.

Cardiac

Moxifloxacin and other FQs to a lesser degree carry a risk of **QTc prolongation** leading to possible arrhythmias, most notably torsades de pointes. This occurs due to inhibition of specific potassium channels. Hence, EKG monitoring is recommended for drugs that have the risk of QT prolongation.

Neurological

Given their generally very good CNS penetration, side effects such as headache, dizziness, **delirium, agitation**, and memory impairment have been reported with just a single use ore likely to cause altered mental status. **Seizures** and **peripheral neuropathy** are rare ADRs. The peripheral neuropathy can last from months to years or be permanent.

What are the known neurologic side effects of FQs?

Dermatological

Allergies and rashes are also somewhat rare but known adverse effects of FQs. Affecting roughly 1% to 2% of patients, rashes are generally mild but do worsen with prolonged treatment. **Photosensitivity and phototoxicity** are also real concerns for some patients, with sunscreen use being a definite recommendation (though with questionable efficacy). More severe hypersensitivity reactions to FQs (such as Stevens-Johnson syndrome and toxic epidermal necrolysis) are rare but have been reported (Figure 4). Patients affected should avoid taking FQs for life.

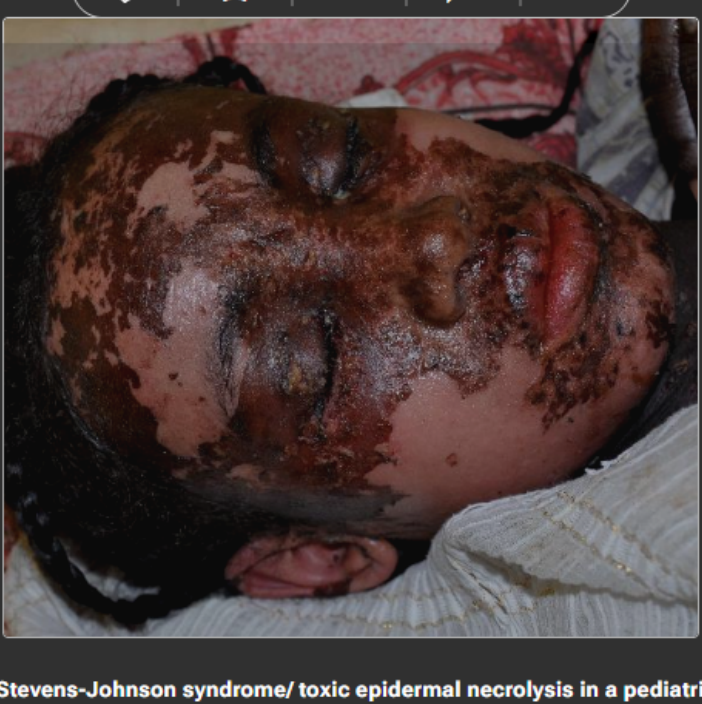


Figure 4 (Stevens-Johnson syndrome/ toxic epidermal necrolysis in a pediatric patient)

Credit: Courtesy of Dr. Richard Usatine.

Musculoskeletal

FQs famously affect the bones and tendons, especially in children. **Cartilage erosion and arthropathy** have been noted in children in weight-bearing joints.

Tendinopathy and tendon rupture are uncommon but real considerations in those being treated with FQs. Most cases involve the Achilles tendon, but other tendons may be affected.

Which tendon is most commonly affected by FQ-induced tendinopathy?

CLINICAL CORRELATION

Since FQs are rarely used in children, it is difficult to gather data on their effects in the pediatric population. However, ciprofloxacin's use in patients with CF in whom death due to *Pseudomonas* infection is a weightier risk than what risks long-term FQ use would confer has provided an accessible source of retrospective data for researchers, but risks/ benefits should be weighed for each patient.

Dysglycemia

FQs have been associated with both hyper- and hypoglycemia. FQs now carry a warning about the risk of hypoglycemia particularly in elderly patients with diabetes.

Aortic aneurysm ar

Use of FQs have been shown to increase the risk of aortic aneurysms or dissection. In patients with risk of aortic aneurysms, it is best to avoid FQs.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to VF, how do you explain this illness to her?

You suspect acute pyelonephritis, and a urine test at the office confirms infection. You explain, "You've got a kidney infection called pyelonephritis. This is more serious than a simple bladder infection; it is making your 'whole-body' sick. We need to treat this right away. I'll prescribe an antibiotic called ciprofloxacin that will cover the most likely bacteria to cause this. We'll also send a urine culture test to verify that your infection is susceptible to this drug: we'll change it if necessary." You brief VF on other measures to treat her pyelonephritis at home and send her to the lab for the urine culture and blood tests, including a CBC and renal function tests.

Summary

- Fluoroquinolones (FQs) are antibiotics that inhibit DNA gyrase and topoisomerase IV, leading to bacterial cell death.
- Resistance to FQs develops through efflux pump modifications and target site alterations, with widespread resistance noted in both gram-positive and gram-negative bacteria.
- FQs are categorized by their use, with levofloxacin and moxifloxacin being key for respiratory infections due to their efficacy against S. pneumoniae.
- FQs are absorbed penetration, but their absorption can be decreased by multivalent cations found in antacids and supplements.
- Common adverse effects of FQs include gastrointestinal disturbances, QTc prolongation, neurologic side effects, and tendinopathy, particularly affecting the Achilles tendon.
- Due to potential adverse effects, FQs are generally not recommended for children.

Review Questions

1. Which tendon is most commonly affected by FQs?

A. ○ Achilles tendon
B. ○ Patellar tendon
C. ○ Biceps tendon
D. ○ Rotator cuff tendon
E. ○ Quadriceps tendon

Explanation (requires correct answer)

2. What is the primary action of fluoroquinolones on DNA?

A. ○ Inhibit typ
B. ○ Blocking RNA polymerase
C. ○ Inhibiting helicase
D. ○ Enhancing supercoiling
E. ○ Binding to DNA

Explanation (requires correct answer)

3. What are the two main mechanisms of bacterial resistance against Fluoroquinolones?

A. ○ Modifications to efflux pumps and alteration of target site
B. ○ Mutation in DNA gyrase
C. ○ Mutation in topoisomerase IV
D. ○ Mutation in ligase domain

Explanation (requires correct answer)

4. What common item etabolism?

A. ○ Multivalent cations
B. ○ Proteins
C. ○ Carbohydrates
D. ○ Fats
E. ● Alcohol

Explanation (requires correct answer)

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 0/4